

Regression Project – Group 3 (Intermediate Report)

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```
suppressPackageStartupMessages({  
  library(tidyverse)  
  library(readxl)  
  library(simFrame)  
  library(dplyr)  
  library(ggplot2)
```

```
library(tidyr)
library(forcats)
library(effects)
library(gt)
library(corrplot)
library(splines)
library(car)
library(ggeffects)
library(GGally)
library(effects)

})
```

1 1. Introduction

Employment income is a central indicator of economic well being, yet it is shaped by a complex interplay of demographic and social factors. In this report, we investigate how gender, citizenship, age, and household size influence income levels in South Austria, focusing specifically on the regions of Carinthia and Styria using the EU SILC panel dataset.

Our analysis proceeds in three stage:

- Data preparation and descriptive statistics to identify distributions, outliers, and basic associations.
- Regression modeling using linear, polynomial, and spline specifications to capture both straightforward and nonlinear relationships.
- Interaction effects to explore how demographic variables combine to produce differentiated income outcomes.

From a methodological perspective, we employ:

- Polynomial regression to model quadratic curvature in age and household size effects.
- Spline regression to flexibly capture turning points and plateaus in continuous predictors.
- Interaction terms to test whether the impact of one variable (e.g., age) depends on another (e.g., gender or citizenship).
- Diagnostic checks to validate model assumptions and ensure robustness.

By combining descriptive visualization with advanced regression techniques, this report provides a nuanced view of income determinants in South Austria. The findings highlight both individual and joint effects of demographic factors, offering insights into inequality patterns and the importance of flexible modeling approaches.

2 2. Data Collection and Preparation

```
data("eusilcP")

dat <- eusilcP %>%
  select(py010n, gender, citizenship, hsize, age, region) %>%
  filter(region %in% c("Carinthia", "Styria")) %>%
  filter(py010n > 0)
dat <- dat %>% rename(income = py010n)
dat$hsize <- as.numeric(as.character(dat$hsize))
dat <- na.omit(dat)
```

3 3. Descriptive Analysis

3.1 3.1 Numeric Summaries

```
num_vars <- dat %>% select(income, age, hsize)
summary(num_vars)
```

income		age		hsize	
Min.	: 1.93	Min.	:16.00	Min.	:1.00
1st Qu.:	10066.01	1st Qu.:	29.00	1st Qu.:	2.00
Median	: 16225.84	Median	:40.00	Median	:3.00
Mean	: 16952.35	Mean	:39.73	Mean	:3.19
3rd Qu.:	21939.78	3rd Qu.:	49.00	3rd Qu.:	4.00
Max.	:118362.27	Max.	:97.00	Max.	:9.00

3.2 3.2 Frequency Tables

```
table(dat$gender)
```

```
male female
3004  2263
```

```
table(dat$citizenship)
```

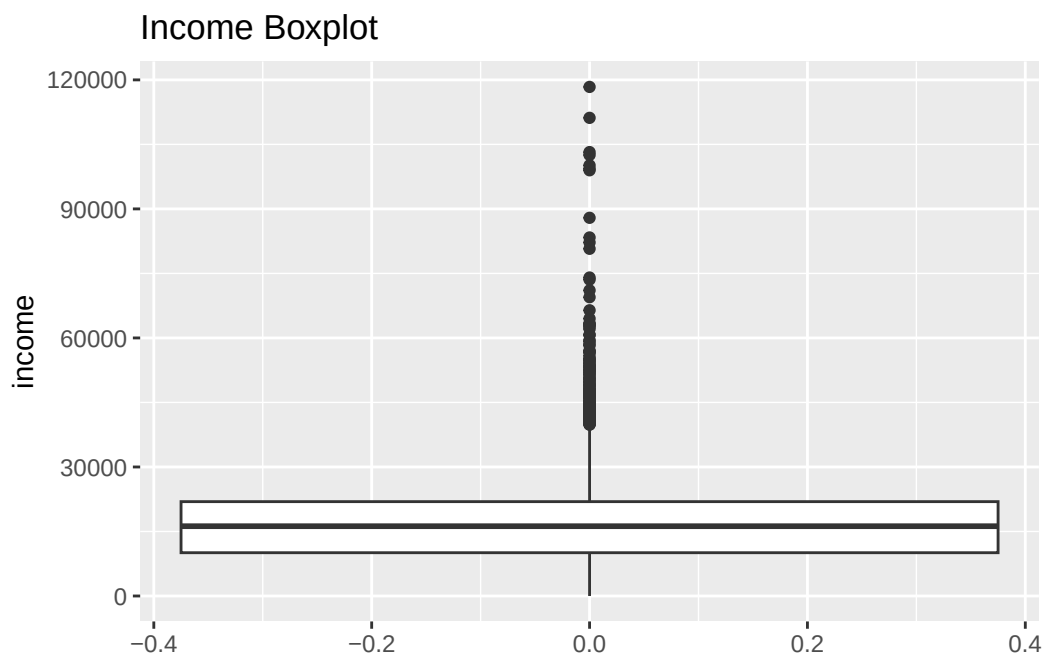
	AT	EU	Other
	5021	79	167

```
table(dat$gender, dat$citizenship)
```

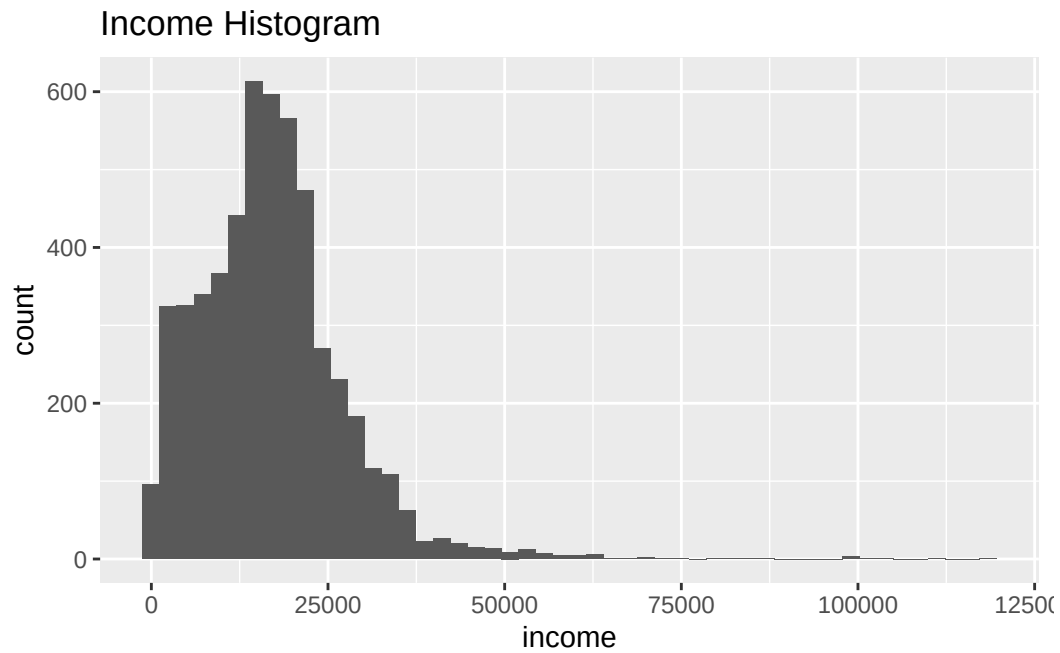
	AT	EU	Other
male	2867	40	97
female	2154	39	70

3.3 3.3 Univariate Plots

```
# Income  
ggplot(dat, aes(y = income)) +  
  geom_boxplot() +  
  labs(title = "Income Boxplot")
```

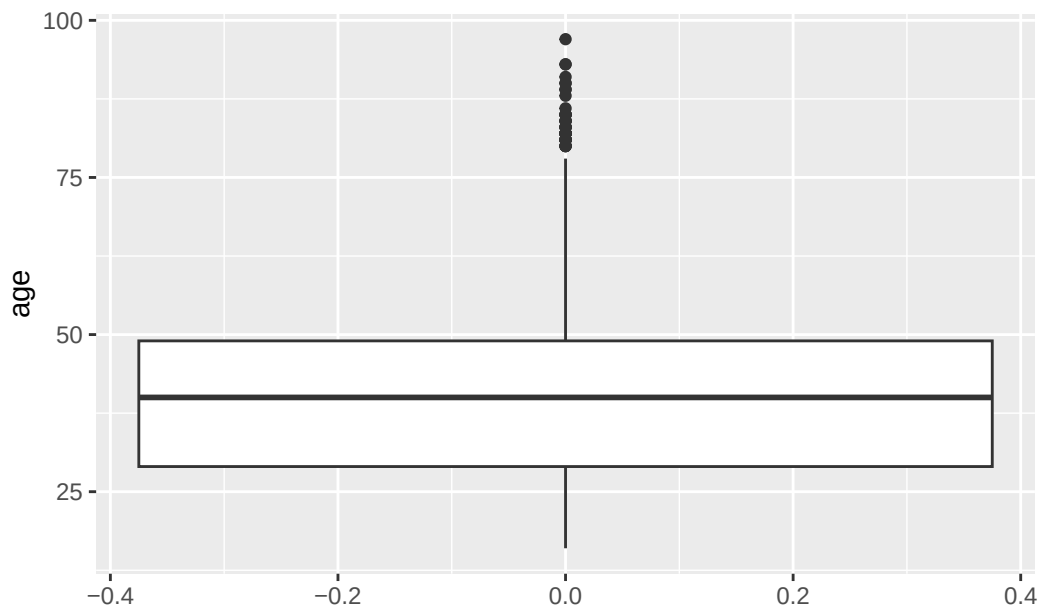


```
ggplot(dat, aes(x = income)) +  
  geom_histogram(bins = 50) +  
  labs(title = "Income Histogram")
```



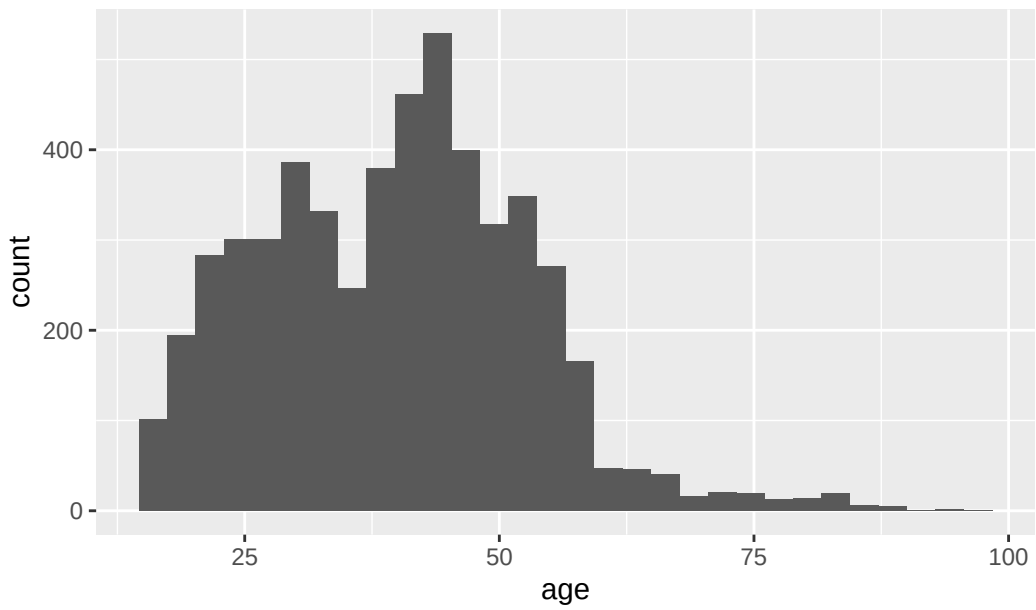
```
# Age  
ggplot(dat, aes(y = age)) +  
  geom_boxplot() +  
  labs(title = "Age Boxplot")
```

Age Boxplot

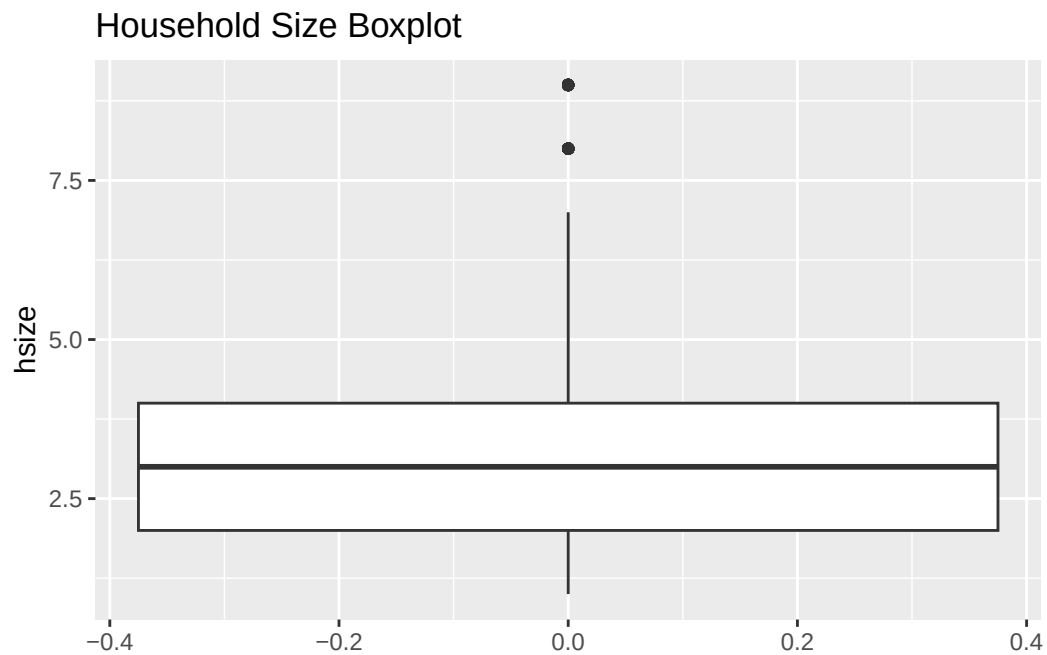


```
ggplot(dat, aes(x = age)) +  
  geom_histogram(bins = 30) +  
  labs(title = "Age Histogram")
```

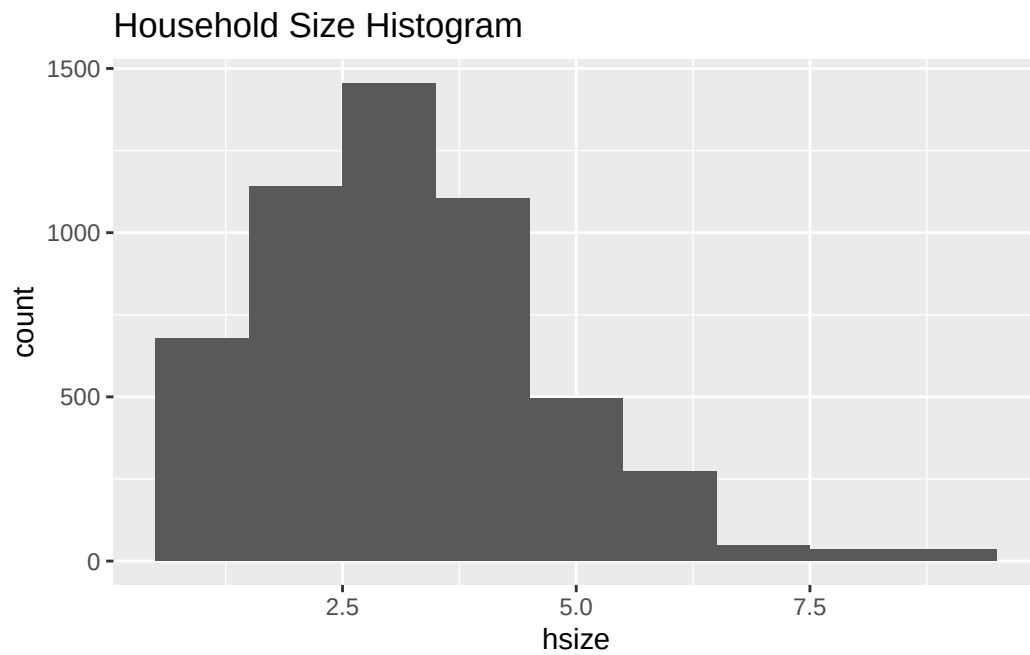
Age Histogram



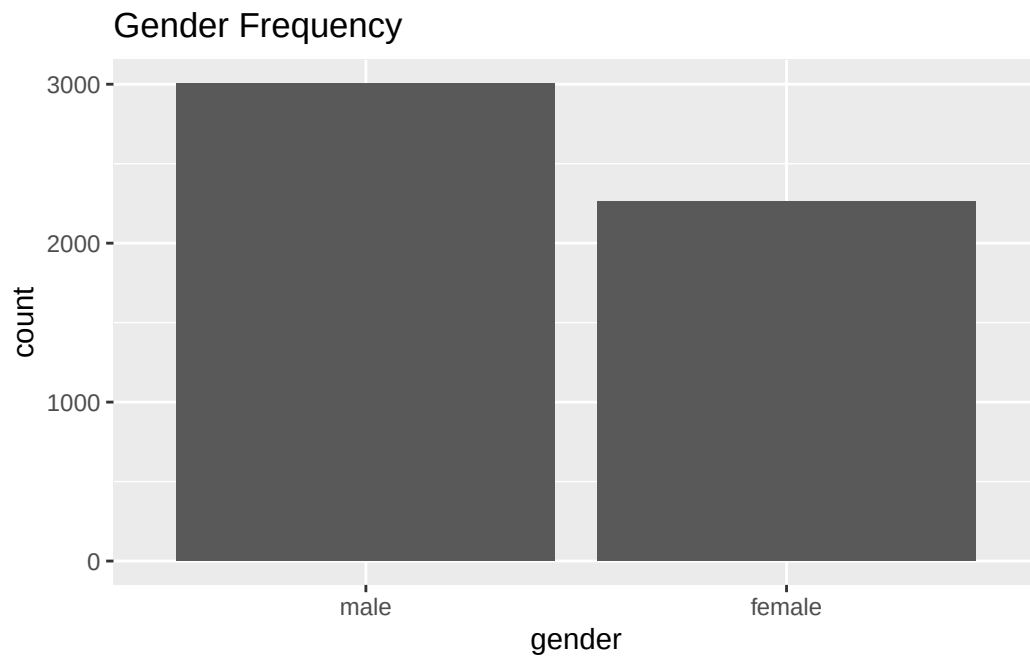
```
# Household Size
ggplot(dat, aes(y = hsize)) +
  geom_boxplot() +
  labs(title = "Household Size Boxplot")
```



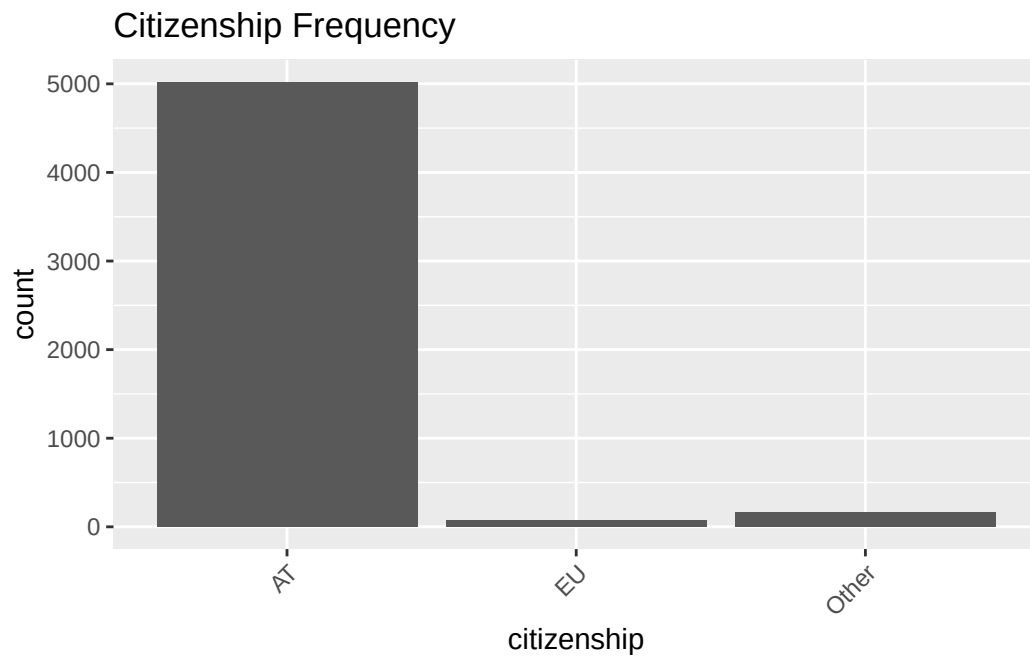
```
ggplot(dat, aes(x = hsize)) +
  geom_histogram(binwidth = 1) +
  labs(title = "Household Size Histogram")
```



```
# Gender
ggplot(dat, aes(x = gender)) +
  geom_bar() +
  labs(title = "Gender Frequency")
```

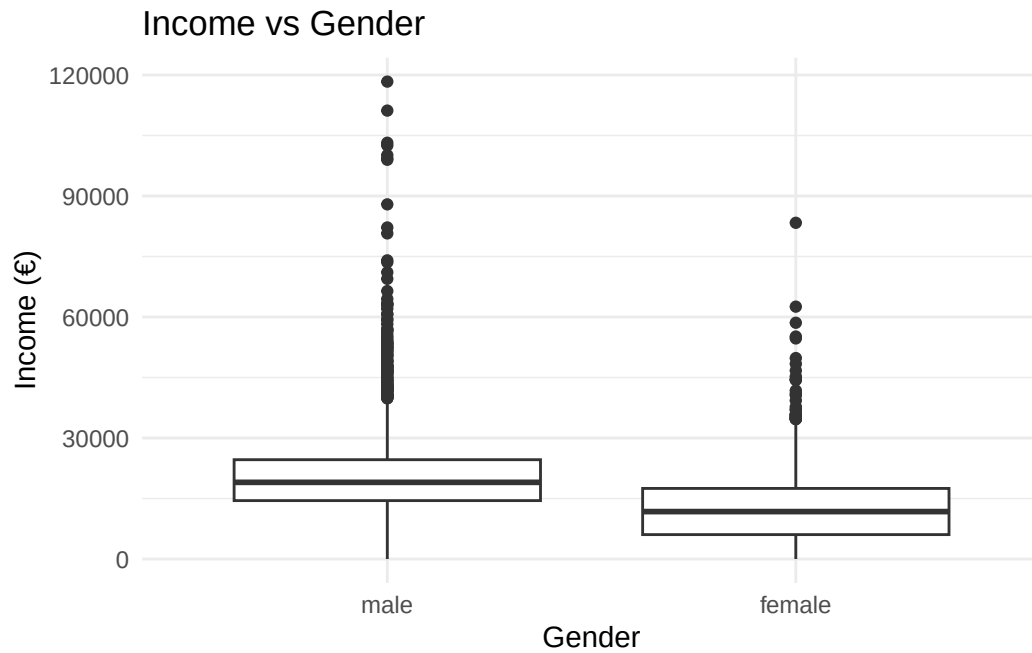


```
# Citizenship  
ggplot(dat, aes(x = citizenship)) +  
  geom_bar() +  
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +  
  labs(title = "Citizenship Frequency")
```

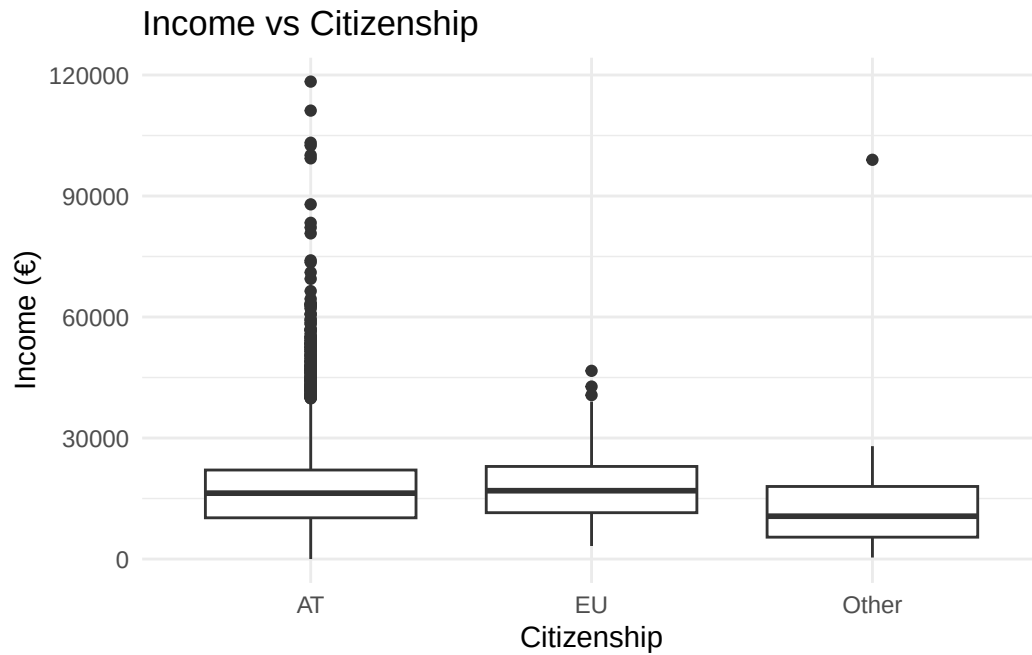


3.4 3.4 Bivariate Plots

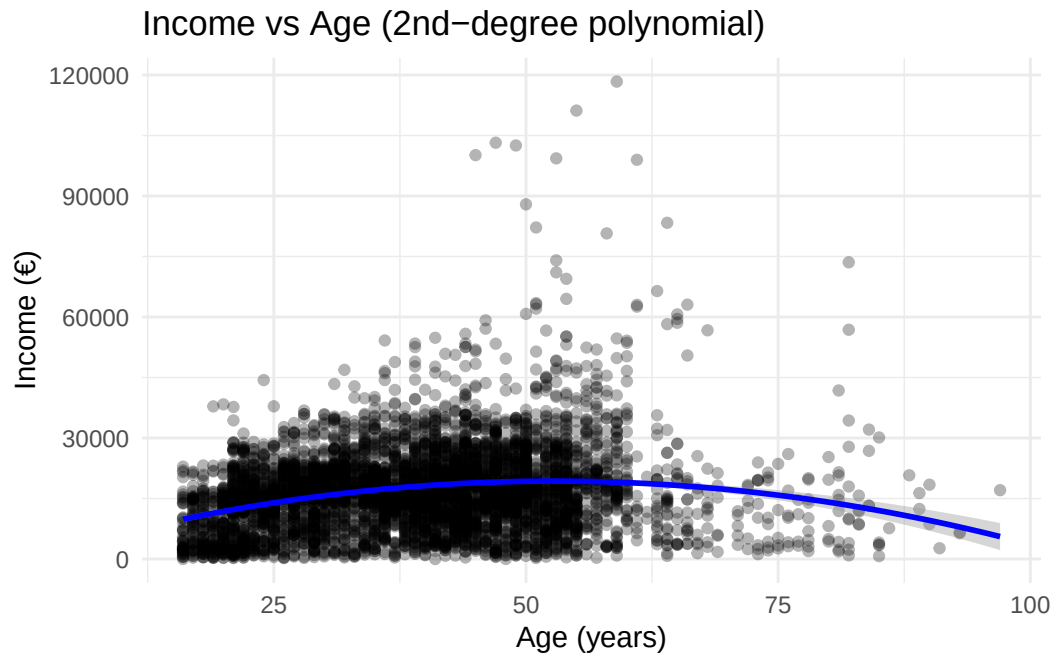
```
# Gender vs Income
ggplot(dat, aes(x = gender, y = income)) +
  geom_boxplot() +
  labs(title = "Income vs Gender", y = "Income (€)", x = "Gender") +
  theme_minimal()
```



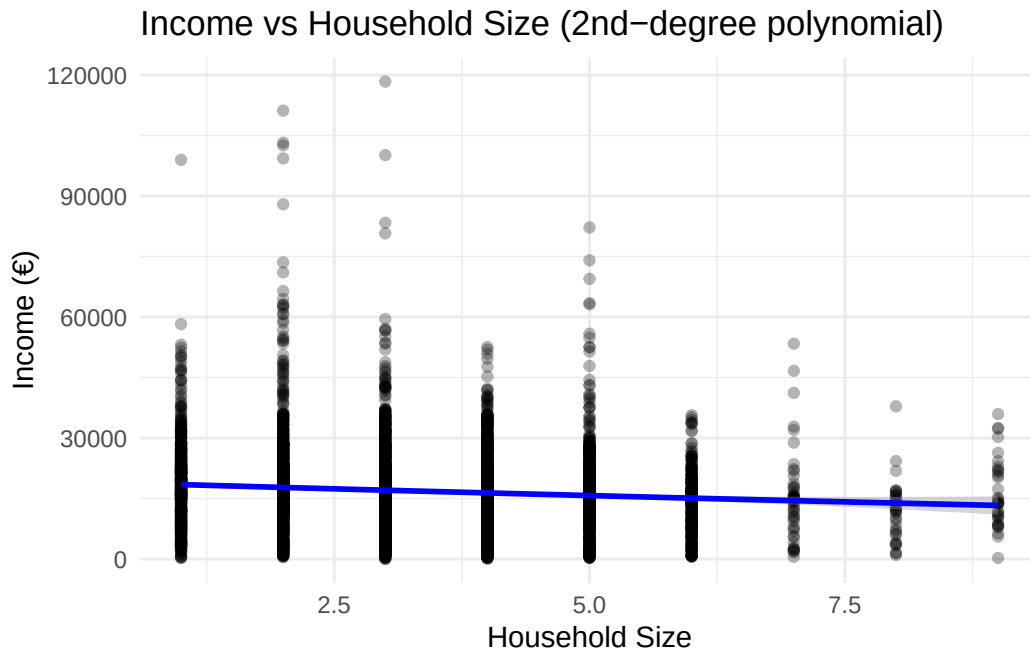
```
# Citizenship vs Income
ggplot(dat, aes(x = citizenship, y = income)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Income vs Citizenship", y = "Income (€)", x = "Citizenship") +
  theme_minimal()
```



```
# Age vs Income (Polynomial)
ggplot(dat, aes(x = age, y = income)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), color = "blue") +
  labs(title = "Income vs Age (2nd-degree polynomial)",
       x = "Age (years)", y = "Income (€)") +
  theme_minimal()
```

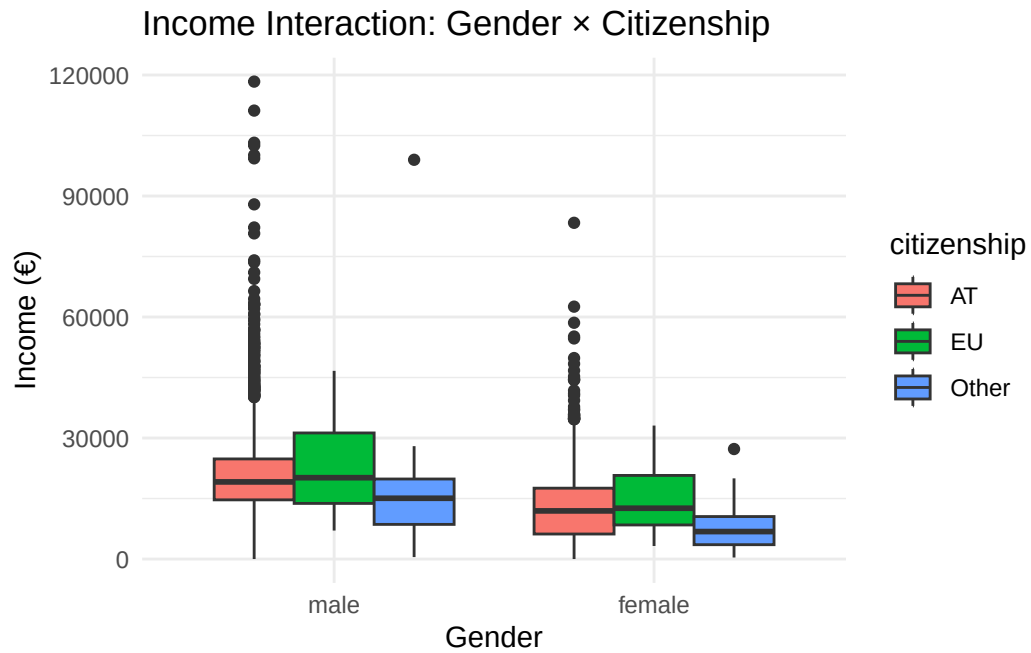


```
# Household size vs Income (Polynomial)
ggplot(dat, aes(x = hsize, y = income)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), color = "blue") +
  labs(title = "Income vs Household Size (2nd-degree polynomial)",
       x = "Household Size", y = "Income (€)") +
  theme_minimal()
```



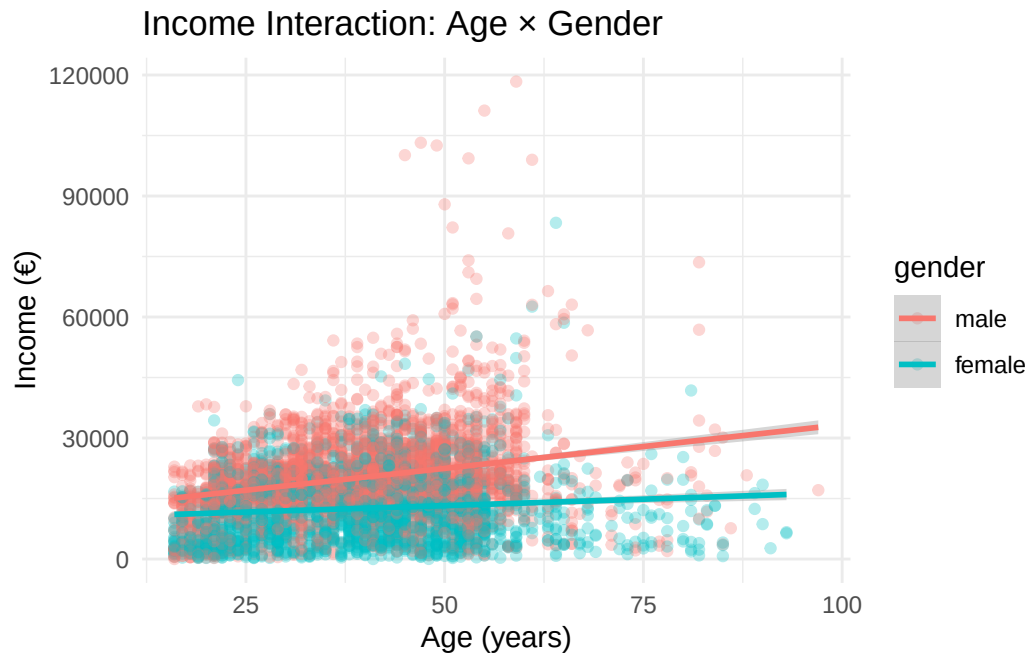
3.5 3.5 Interaction Plots

```
# Gender × Citizenship
ggplot(dat, aes(x = gender, y = income, fill = citizenship)) +
  geom_boxplot(position = "dodge") +
  labs(title = "Income Interaction: Gender × Citizenship",
       y = "Income (€)", x = "Gender") +
  theme_minimal()
```



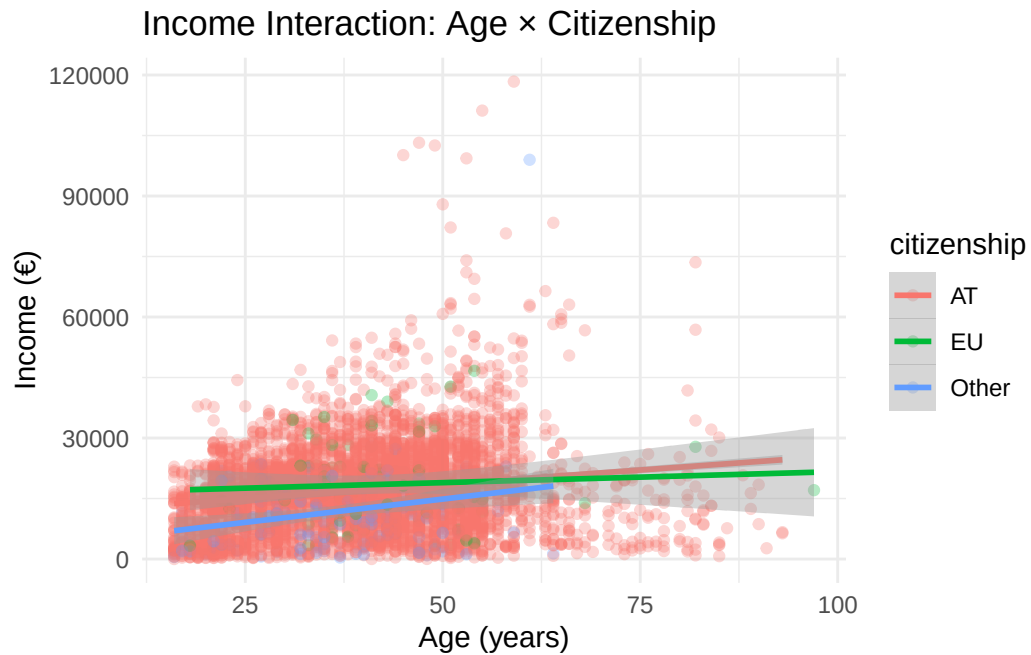
```
# Age × Gender
ggplot(dat, aes(x = age, y = income, color = gender)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm") +
  labs(title = "Income Interaction: Age × Gender",
       x = "Age (years)", y = "Income (€)") +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'



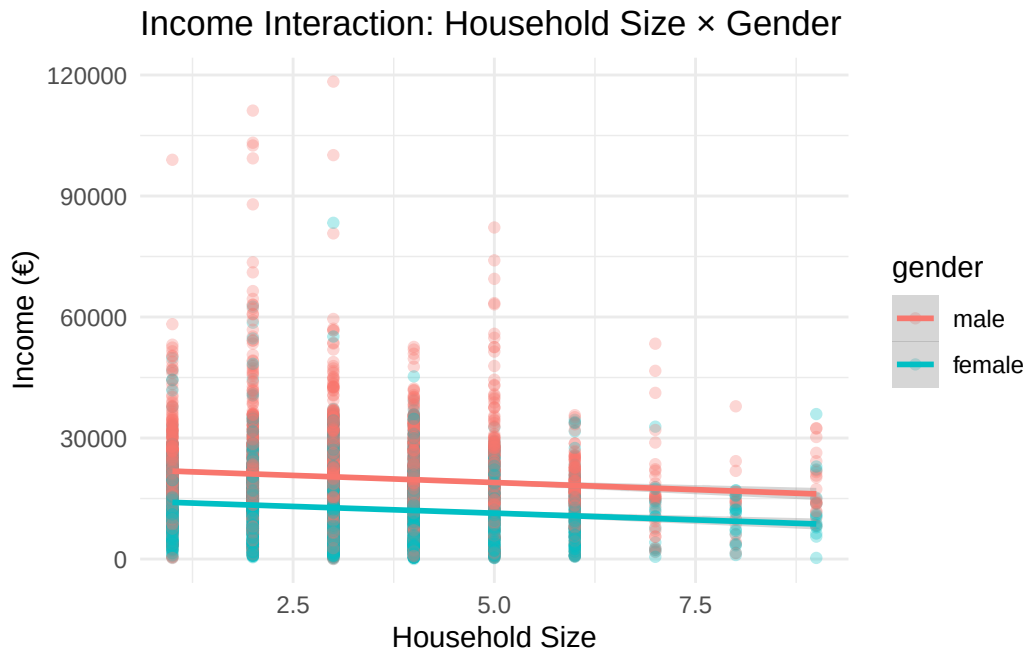
```
# Age × Citizenship
ggplot(dat, aes(x = age, y = income, color = citizenship)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm") +
  labs(title = "Income Interaction: Age × Citizenship",
       x = "Age (years)", y = "Income (€)") +
  theme_minimal()
```

``geom_smooth()`` using formula = 'y ~ x'



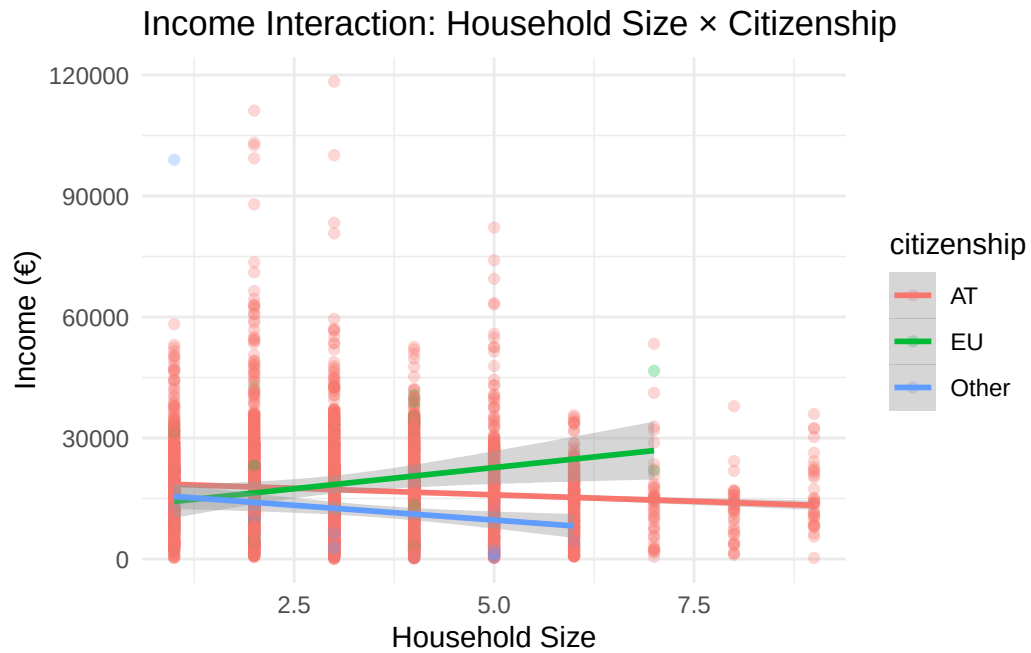
```
# Household Size × Gender
ggplot(dat, aes(x = hsize, y = income, color = gender)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm") +
  labs(title = "Income Interaction: Household Size × Gender",
       x = "Household Size", y = "Income (€)") +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'



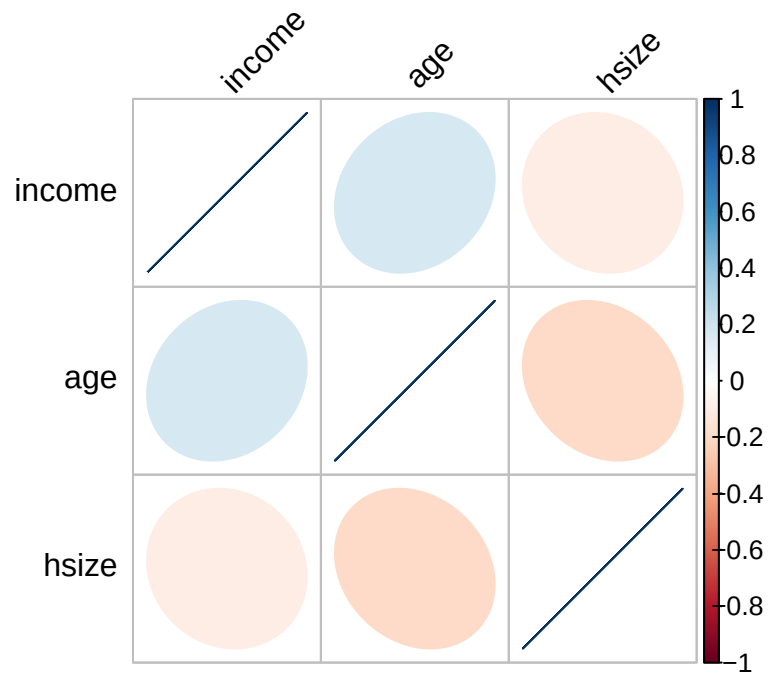
```
# Household Size × Citizenship
ggplot(dat, aes(x = hsize, y = income, color = citizenship)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm") +
  labs(title = "Income Interaction: Household Size × Citizenship",
       x = "Household Size", y = "Income (€)") +
  theme_minimal()
```

``geom_smooth()`` using formula = 'y ~ x'



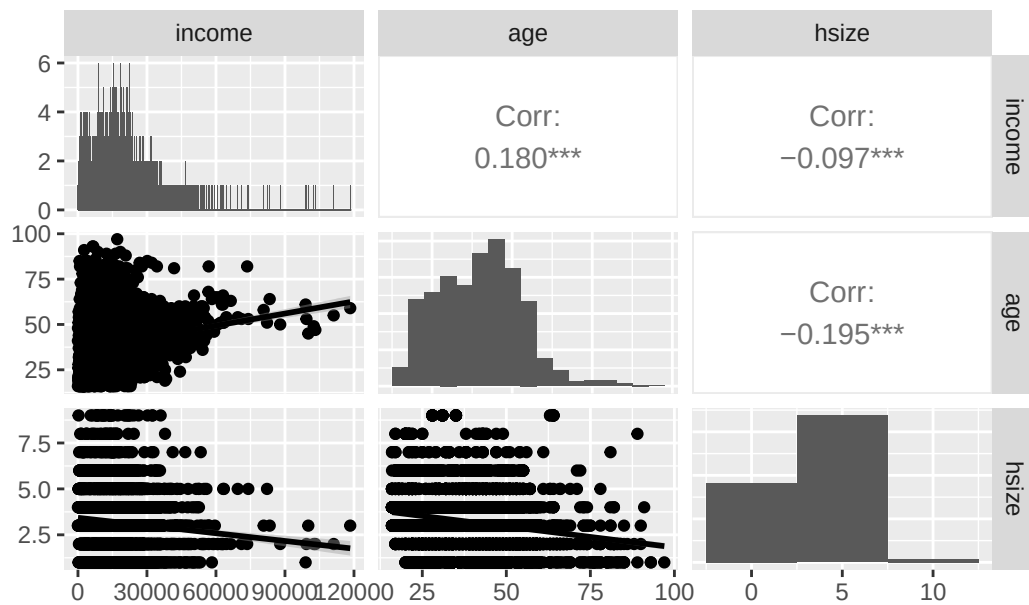
3.6 3.6 Correlation / Scatterplot Matrix

```
# Correlation matrix
cor_mat <- cor(num_vars)
corrplot(cor_mat, method = "ellipse", tl.col = "black", tl.srt = 45)
```



```
# Scatterplot matrix (modern version)
ggpairs(num_vars,
  title = "Scatterplot Matrix of Income, Age, Household Size",
  upper = list(continuous = "cor"),
  lower = list(continuous = "smooth"),
  # Option A: histograms with custom binwidth
  diag = list(continuous = wrap("barDiag", binwidth = 5)))
```

Scatterplot Matrix of Income, Age, Household Size



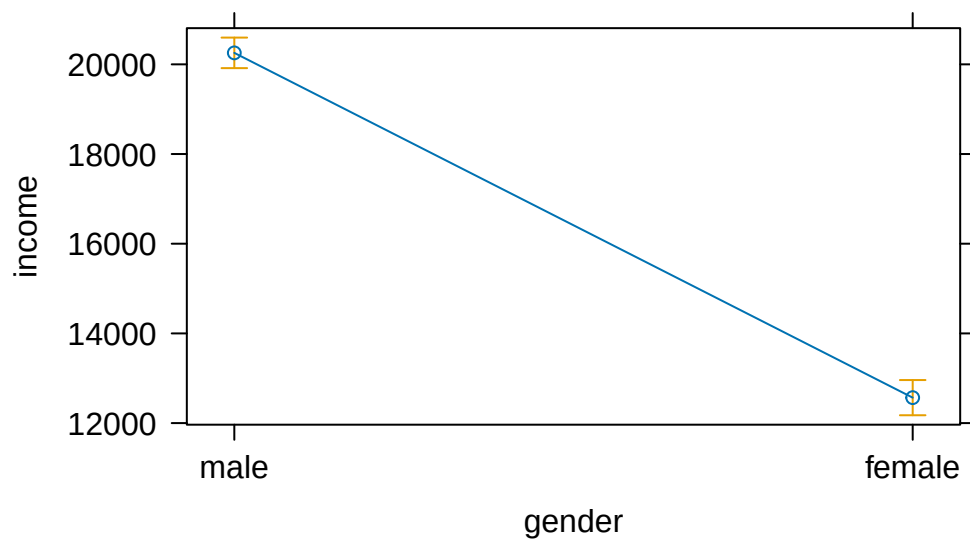
4 4. Regression Modeling

4.1 4.1 Linear Model

```
# Define linear model with renamed variable
lm_model <- lm(income ~ gender + citizenship + hsize + age, data = dat)

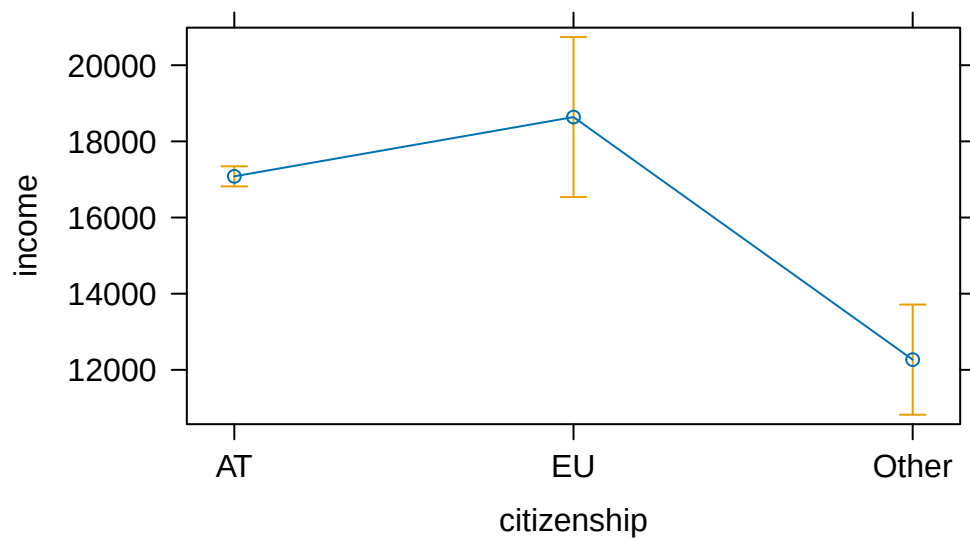
# Gender effect
plot(Effect("gender", lm_model),
     main = "Linear Effect: Gender")
```

Linear Effect: Gender

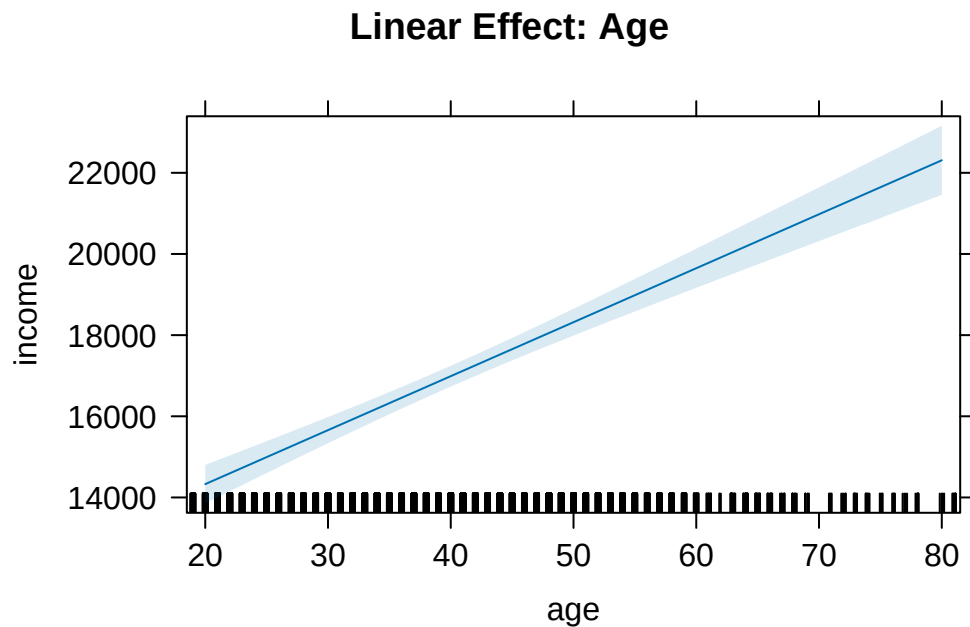


```
# Citizenship effect  
plot(Effect("citizenship", lm_model),  
     main = "Linear Effect: Citizenship")
```

Linear Effect: Citizenship

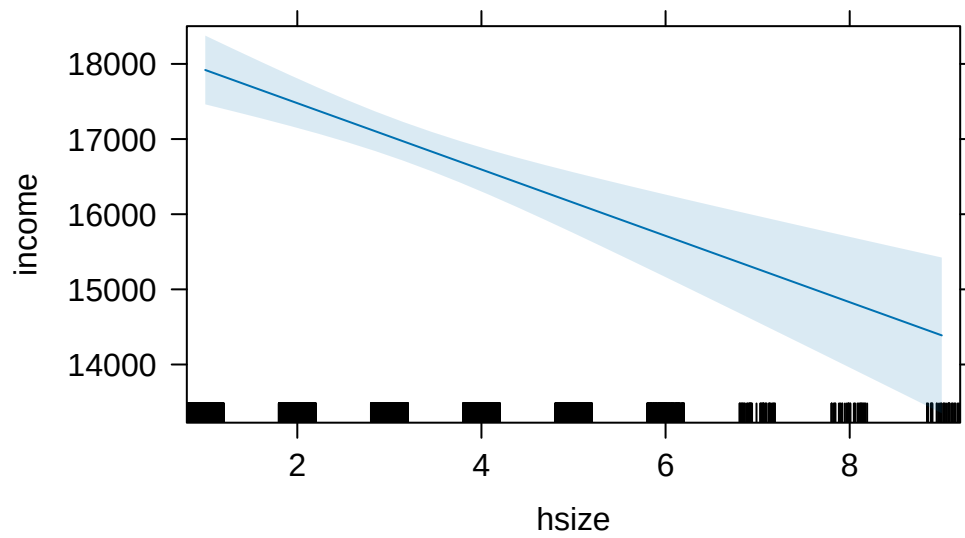


```
# Age effect (linear)
plot(Effect("age", lm_model,
          xlevels = list(age = seq(20, 80, by = 5))),
     main = "Linear Effect: Age")
```



```
# Household size effect (linear)
plot(Effect("hsize", lm_model,
          xlevels = list(hsize = 1:9)),
     main = "Linear Effect: Household Size")
```

Linear Effect: Household Size

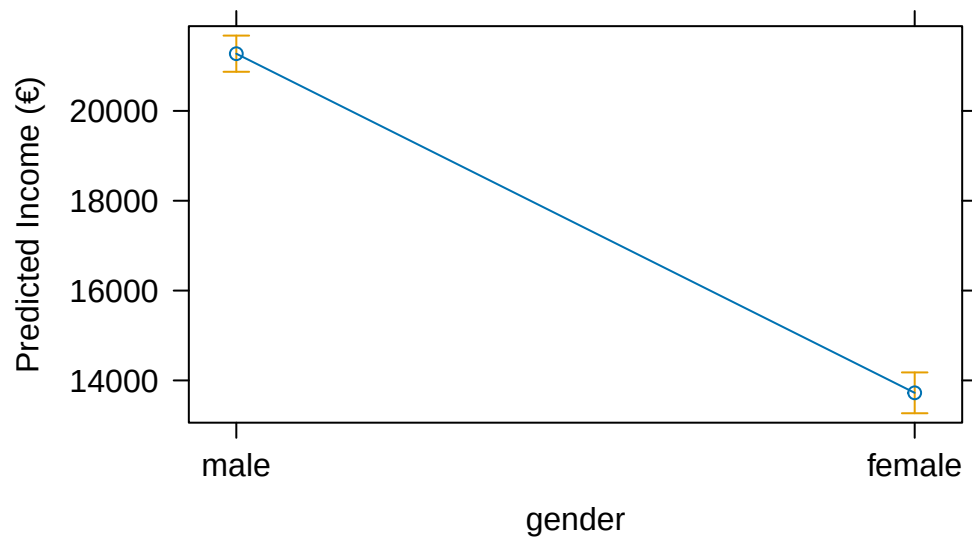


4.2 4.2 Polynomial Model

```
# Define polynomial model with renamed variable
poly_model <- lm(income ~ poly(age, 2) + poly(hsize, 2) + gender + citizenship,
                 data = dat)

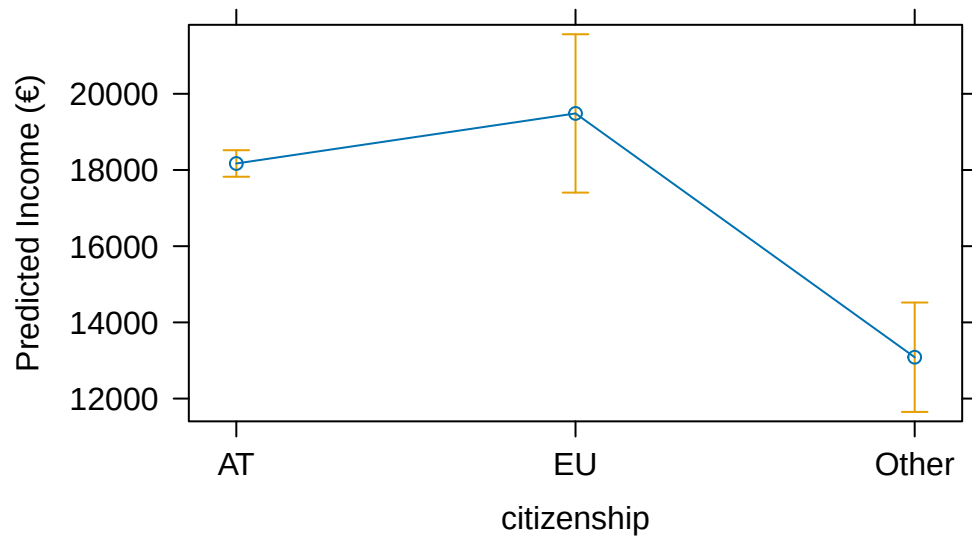
# Gender effect
plot(Effect("gender", poly_model),
     main = "Polynomial Effect: Gender", ylab = "Predicted Income (€)")
```

Polynomial Effect: Gender

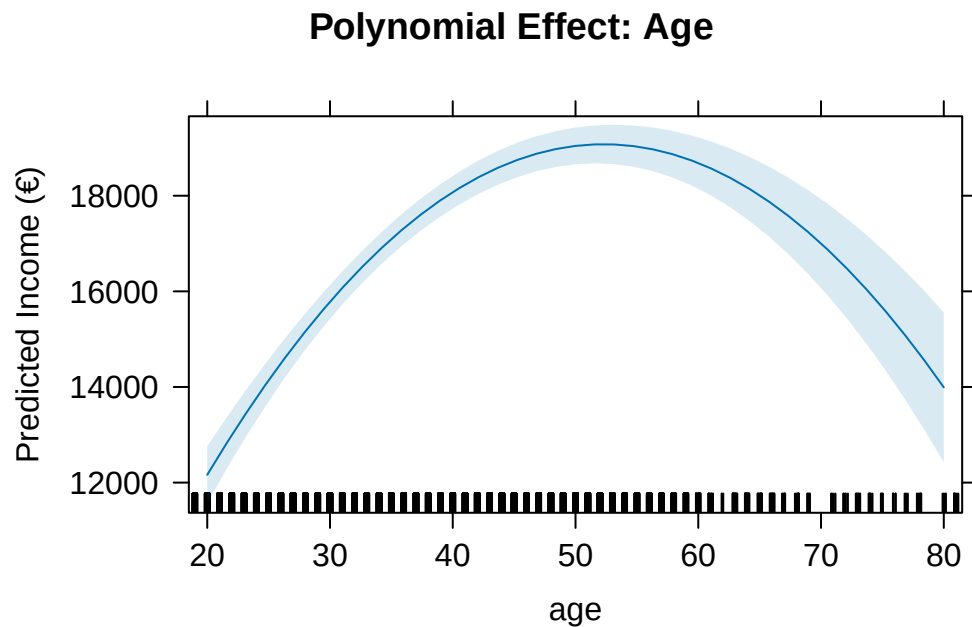


```
# Citizenship effect
plot(Effect("citizenship", poly_model),
     main = "Polynomial Effect: Citizenship", ylab = "Predicted Income (€)")
```

Polynomial Effect: Citizenship

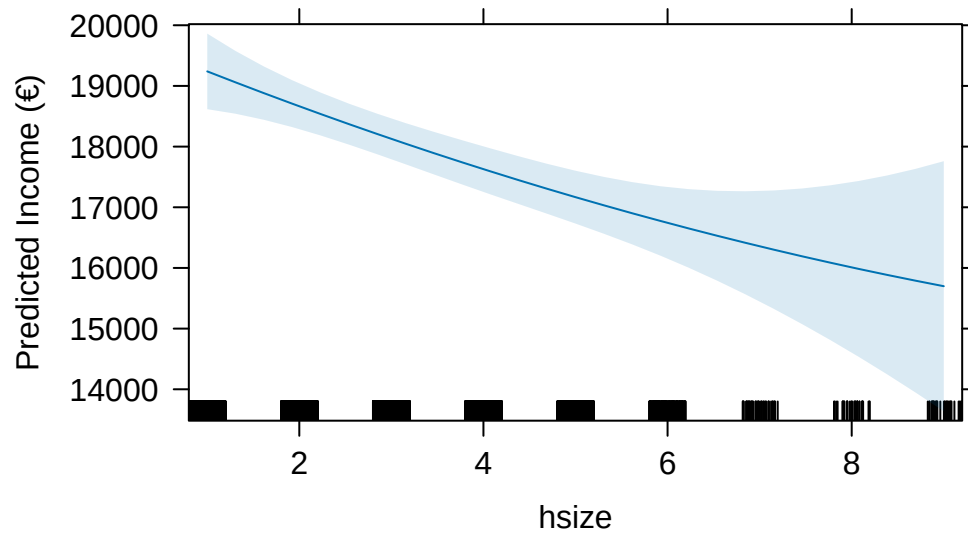


```
# Age effect (quadratic)
plot(Effect("age", poly_model,
          xlevels = list(age = seq(20, 80, by = 5))),
     main = "Polynomial Effect: Age", ylab = "Predicted Income (€)")
```



```
# Household size effect (quadratic)
plot(Effect("hsize", poly_model,
          xlevels = list(hsize = 1:9)),
     main = "Polynomial Effect: Household Size", ylab = "Predicted Income (€)")
```

Polynomial Effect: Household Size



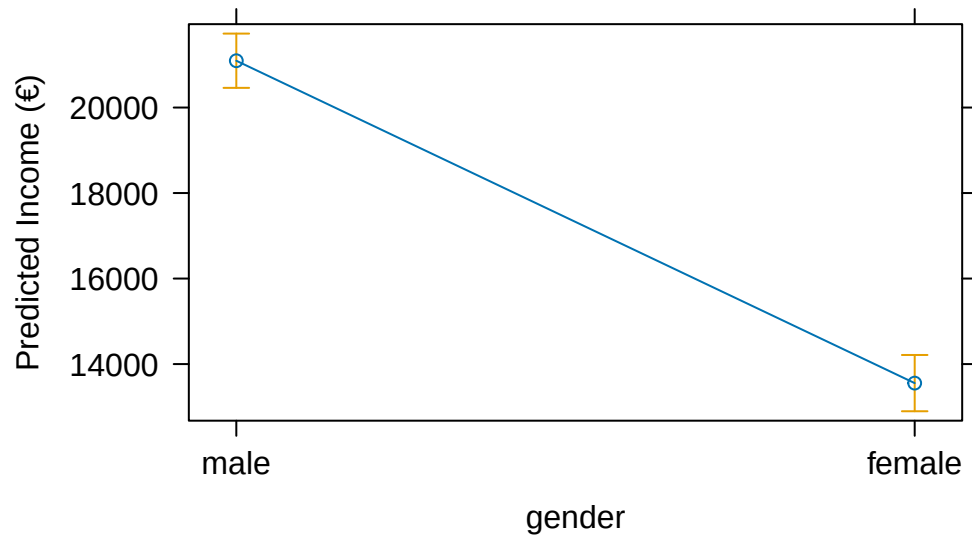
4.3 Spline Model

```
# Define spline model with renamed variable

spline_model <- lm(income ~ ns(age, df = 4) + ns(hsize, df = 4) + gender + citizenship,
                   data = dat)

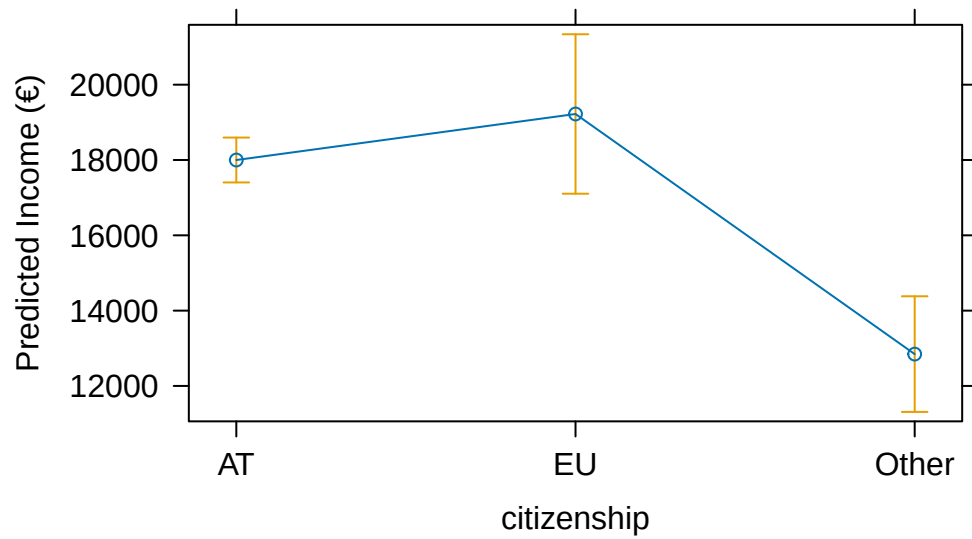
# Gender effect
plot(Effect("gender", spline_model),
     main = "Spline Effect: Gender", ylab = "Predicted Income (€)")
```

Spline Effect: Gender

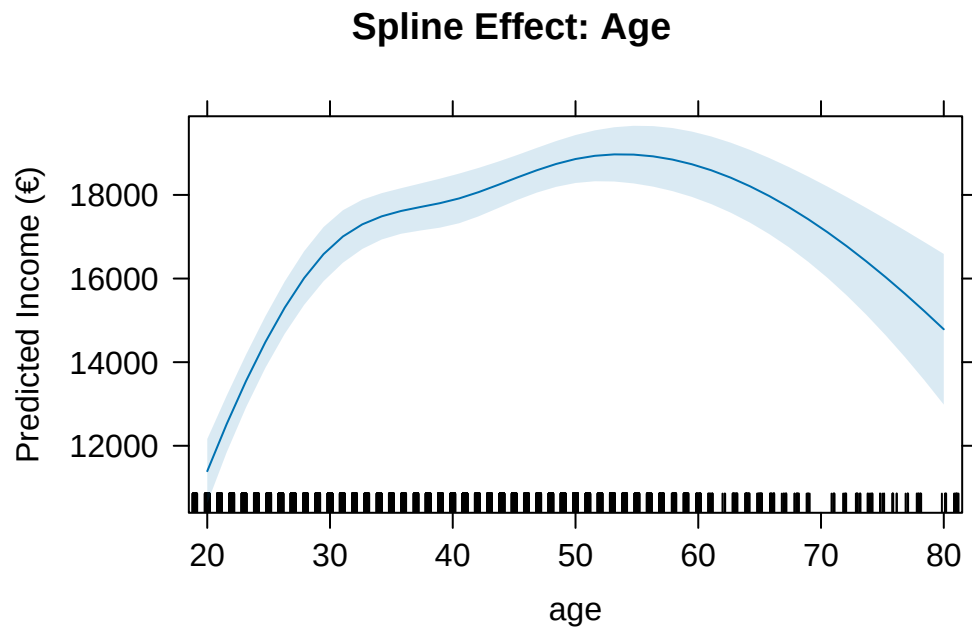


```
# Citizenship effect  
plot(Effect("citizenship", spline_model),  
     main = "Spline Effect: Citizenship", ylab = "Predicted Income (€)")
```

Spline Effect: Citizenship

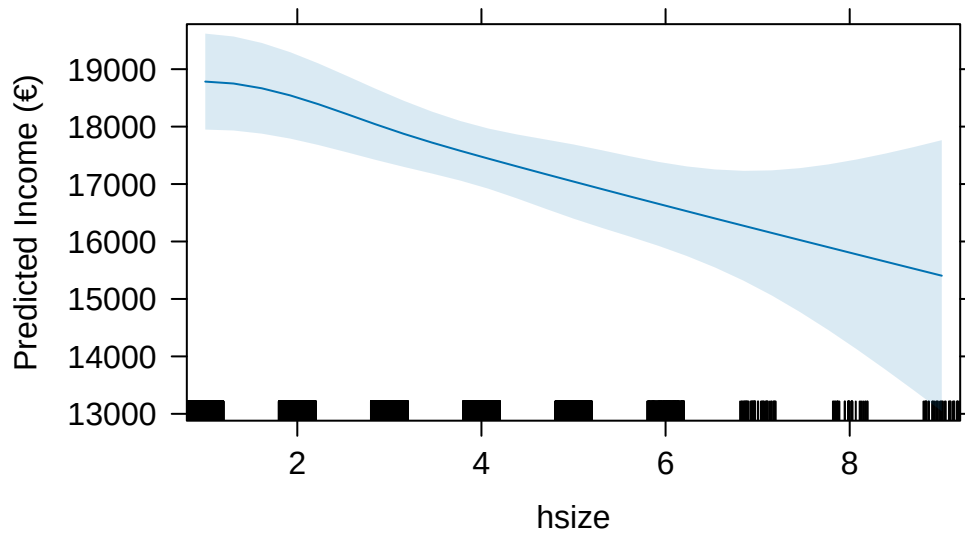


```
# Age effect (spline)
plot(Effect("age", spline_model,
          xlevels = list(age = seq(20, 80, by = 5))),
     main = "Spline Effect: Age", ylab = "Predicted Income (€)")
```



```
# Household size effect (spline)
plot(Effect("hsize", spline_model,
          xlevels = list(hsize = 1:9)),
     main = "Spline Effect: Household Size", ylab = "Predicted Income (€)")
```

Spline Effect: Household Size



4.4 4.4 Interaction Effects

```
# Define interaction model with renamed variable

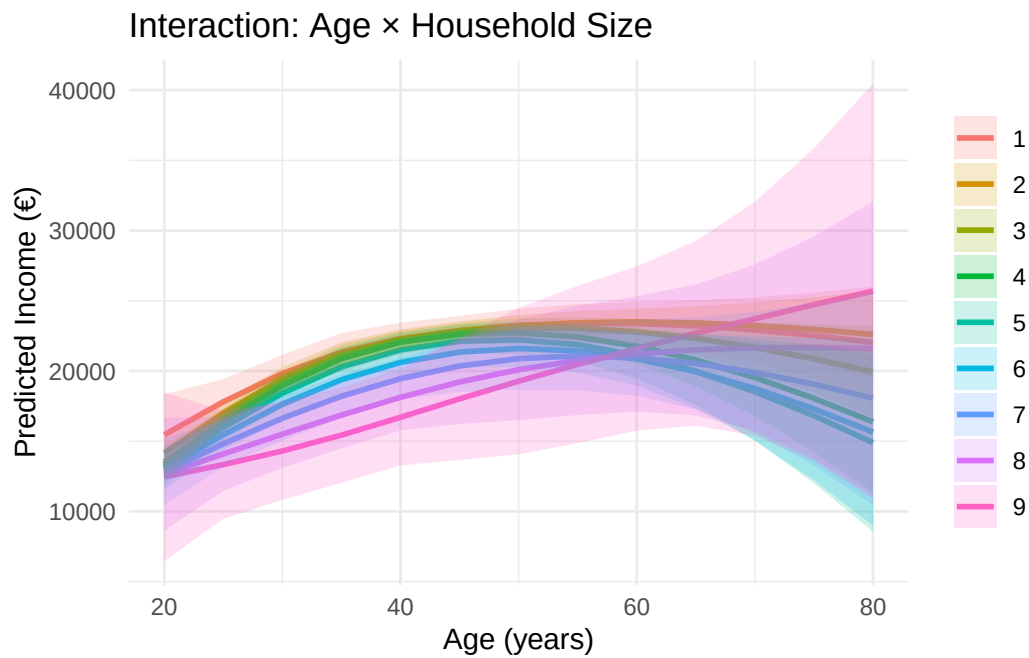
int_model <- lm(income ~ (gender + citizenship + ns(age, 3) + ns(hsize, 3))^2,
               data = dat)

# Age × Household Size interaction using ggeffects
eff_age_hsize <- ggpredict(int_model,
                           terms = c("age [20:80 by=5]", "hsize [1:9]"))

ggplot(eff_age_hsize, aes(x, predicted, color = group)) +
  geom_line(size = 1) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high, fill = group),
            alpha = 0.2, color = NA) +
  labs(title = "Interaction: Age × Household Size",
       x = "Age (years)", y = "Predicted Income (€)") +
  theme_minimal() +
  theme(legend.title = element_blank())
```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.

i Please use ``linewidth`` instead.



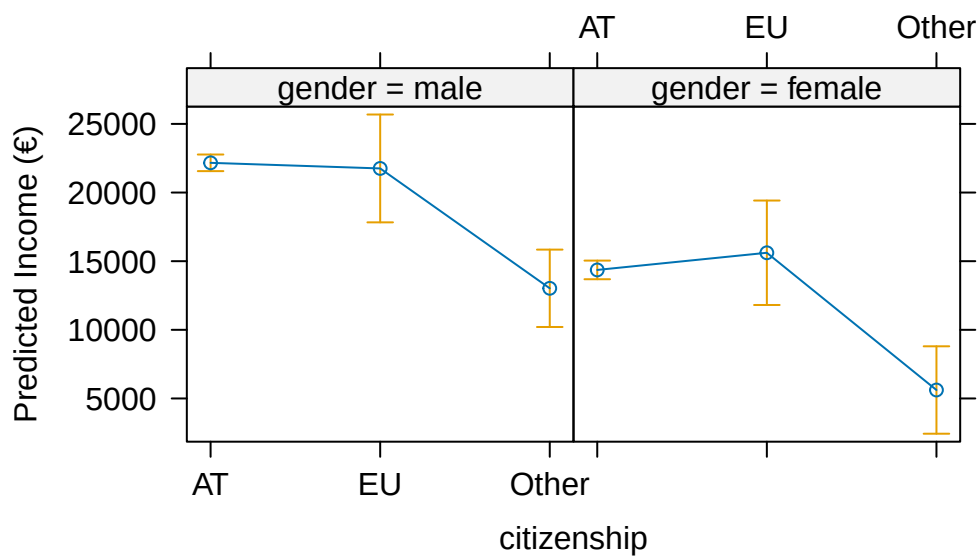
4.5 4.5 Interaction Plots

```
# 4.5 Interaction Plots using effects

int_model <- lm(income ~ (gender + citizenship + ns(age, 3) + ns(hsize, 3))^2,
               data = dat)

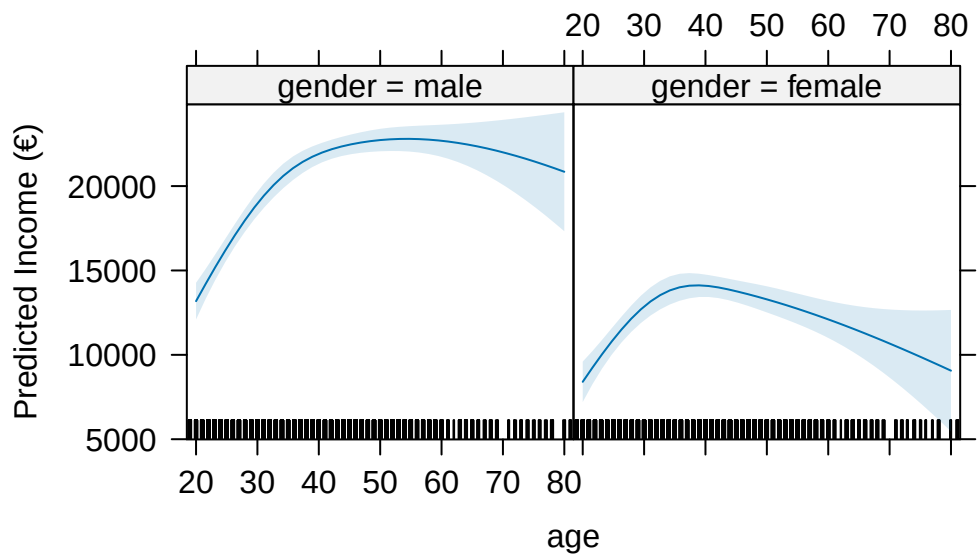
# Interaction plots
plot(Effect(c("gender", "citizenship"), int_model),
     main = "Interaction: Gender × Citizenship", ylab = "Predicted Income (€)")
```

Interaction: Gender × Citizenship

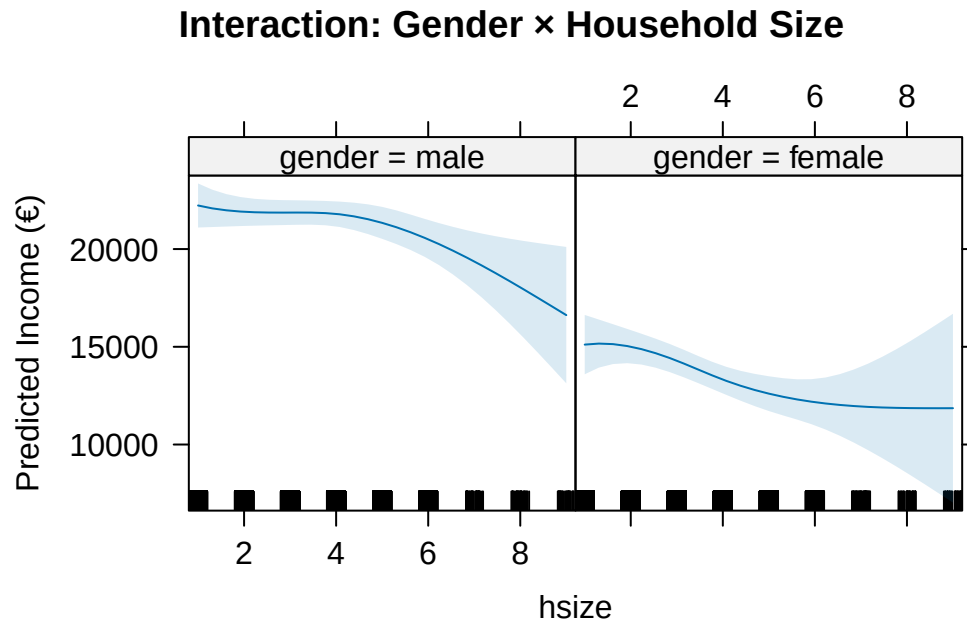


```
plot(Effect(c("gender", "age"), int_model,
  xlevels = list(age = seq(20, 80, by = 5))),
  main = "Interaction: Gender × Age", ylab = "Predicted Income (€)")
```

Interaction: Gender × Age

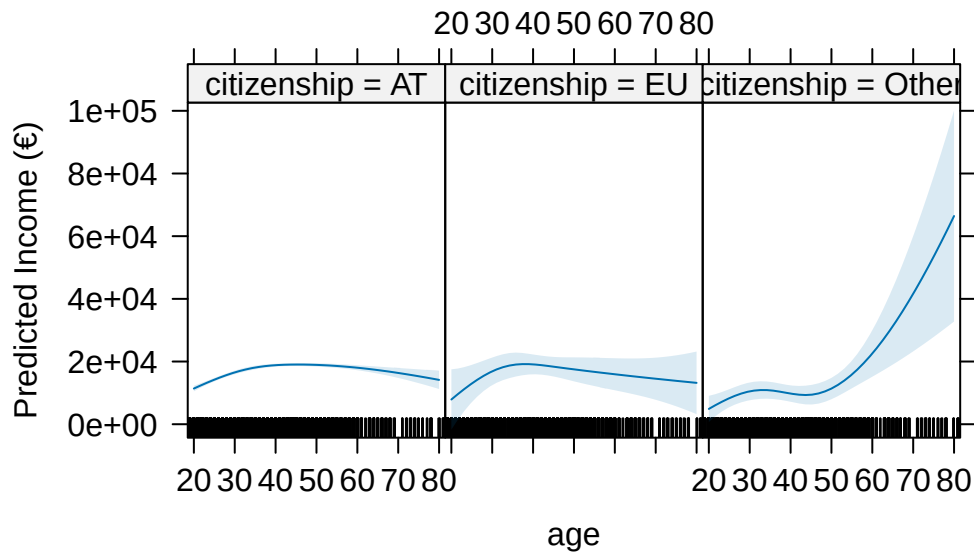


```
plot(Effect(c("gender", "hsize"), int_model,
           xlevels = list(hsize = 1:9)),
     main = "Interaction: Gender × Household Size", ylab = "Predicted Income (€)")
```



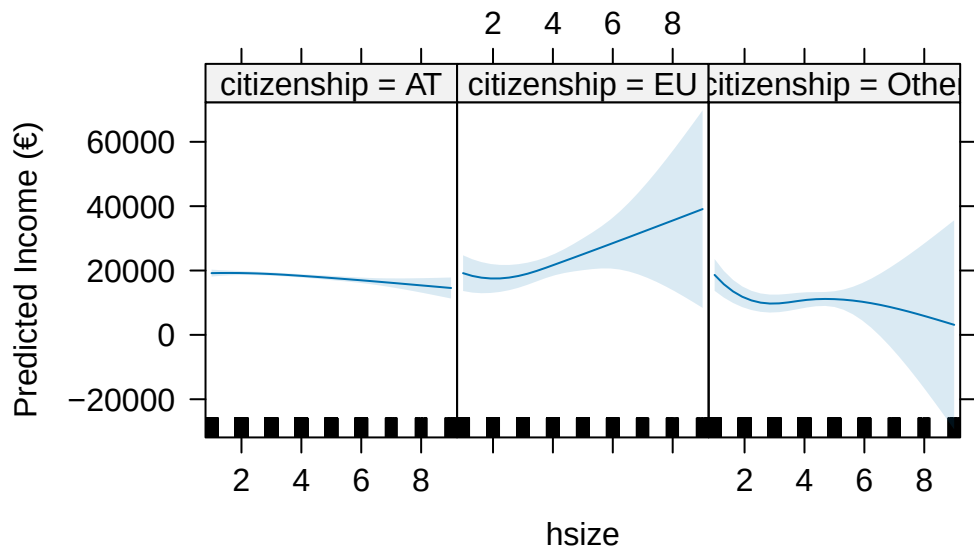
```
plot(Effect(c("citizenship", "age"), int_model,
           xlevels = list(age = seq(20, 80, by = 5))),
     main = "Interaction: Citizenship × Age", ylab = "Predicted Income (€)")
```

Interaction: Citizenship × Age



```
plot(Effect(c("citizenship", "hsize"), int_model,
           xlevels = list(hsize = 1:9)),
     main = "Interaction: Citizenship × Household Size", ylab = "Predicted Income (€)")
```

Interaction: Citizenship × Household Size



5 5. Summary

Income in South Austria is highly skewed, with clear differences by gender and citizenship. Age and household size show nonlinear effects, best captured by polynomial and spline models. Interaction analyses reveal that demographic factors combine in complex ways to shape income outcomes. Overall, flexible regression approaches provide a more accurate picture than simple linear models, and diagnostics confirm model adequacy